Homework 1

September 30, 2026

Some of the problems are collected from the examples in the book.

Problem 1

Let $\mathcal{C}_{n-1}(t)$ denote the set of all continuous polynomial functions defined on $[0, n-1]$ with constant slope in $[i-1, i]$, $i = 1, 2, \ldots, n-1$, n is an integer. With a vector $\alpha(t) \in \mathcal{C}_{n-1}(t)$, we associate an n-dimensional column vector $\alpha = [\alpha(0)\ \alpha(1)\ \ldots\ \alpha(n-1)]^T$.

(1) Show that the vectors $\alpha_i(t) \in \mathcal{C}_{n-1}(t)$, $i = 1, 2, \ldots, n$, are linearly independent if and only if so are the associated column vectors $\alpha_i \in \mathcal{R}^n$, $i = 1, 2, \ldots, n$.

(2) Show that $\alpha_i(t) \in \mathcal{C}_3(t)$, $i = 1, 2, 3, 4$ with

$$\alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad \alpha_4 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

are linearly independent thus form a basis for $\mathcal{C}_3(t)$ and so do the vectors $\beta_i(t) \in \mathcal{C}_3(t)$, $i = 1, 2, 3, 4$ with

$$\beta_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \beta_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \beta_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \beta_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

(3) Find the transition matrix T that transforms the basis $[\alpha_1(t)\ \alpha_2(t)\ \alpha_3(t)\ \alpha_4(t)]$ to the basis $[\beta_1(t)\ \beta_2(t)\ \beta_3(t)\ \beta_4(t)]$.

(4) Find the representation of $\xi(t)$ with

$$\xi = [\xi(0)\ \xi(1)\ \xi(2)\ \xi(3)]^T$$

with respect to the basis $[\beta_1(t)\ \beta_2(t)\ \beta_3(t)\ \beta_4(t)]$.

Problem 2

Please show that the vectors

$$A_1 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

are linearly independent and form a basis for $\mathcal{C}^{2\times 2}$, and find the presentation for the given matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

with respect to this basis.

Problem 3

Let $\mathcal{C}_{n-1}(t)$ denote the set of all continuous polynomial functions defined on $[0, n-1]$ with constant slope in $[i-1, i]$, $i = 1, 2, \ldots, n-1$, n is an integer. Let $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ be a set of vectors of $\mathcal{V}$, the linear span of $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ is referred to as the set of all possible linear representations of these vectors

$$\text{span}\{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s\} = \{\boldsymbol{\alpha} : \boldsymbol{\alpha} = k_1\boldsymbol{\alpha}_1 + k_2\boldsymbol{\alpha}_2 + \cdots + k_s\boldsymbol{\alpha}_s, \quad k_i \in \mathcal{F}\}$$

In this problem, let $\alpha_1(t)$ and $\alpha_2(t) \in \mathcal{C}_3(t)$ with

$$\alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathcal{U} = \text{span}\{\alpha_1(t), \alpha_2(t)\},$$

$\beta_1(t)$ and $\beta_2(t) \in \mathcal{C}_3(t)$ with

$$\beta_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \beta_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathcal{V} = \text{span}\{\beta_1(t), \beta_2(t)\}.$$

(1) The sum of $\mathcal{U}$ and $\mathcal{V}$ is defined as

$$\mathcal{U} + \mathcal{V} = \{\boldsymbol{\alpha} = \boldsymbol{\alpha}_1 + \boldsymbol{\alpha}_2 : \boldsymbol{\alpha}_1 \in \mathcal{U} \text{ and } \boldsymbol{\alpha}_2 \in \mathcal{V}\}.$$

Please find out a basis for $\mathcal{U} + \mathcal{V}$ and its dimension;

(2) The intersect of $\mathcal{U}$ and $\mathcal{V}$ is defined as

$$\mathcal{U} \cap \mathcal{V} = \{\boldsymbol{\alpha} : \boldsymbol{\alpha} \in \mathcal{U} \text{ and } \boldsymbol{\alpha} \in \mathcal{V}\}.$$

Please find out a basis for $\mathcal{U} \cap \mathcal{V}$ and its dimension.

Examples from the Textbook

Read the following examples before attempting Problem 4.

Example 1.23 (p. 21). Let $\mathscr{D} : \mathcal{R}_{n+1}[\lambda] \to \mathcal{R}_n[\lambda]$ be defined as

$$\mathscr{D}(f(\lambda)) = \frac{d}{d\lambda} f(\lambda) \qquad \forall\, f(\lambda) \in \mathcal{R}_{n+1}[\lambda].$$

Then from the linearity of derivative, it is readily to check that $\mathscr{D}$ is a linear transformation.

Example 1.24 (p. 22). Let $\mathscr{S} : \mathcal{R}_n[\lambda] \to \mathcal{R}_{n+1}[\lambda]$ be defined as

$$\mathscr{S}(f(\lambda)) = \int_0^\lambda f(\xi)\, d\xi \qquad \forall\, f(\lambda) \in \mathcal{R}_n[\lambda].$$

Then from the linearity of integration, it is readily to check that $\mathscr{S}$ is a linear transformation.

Example 1.26 (p. 25). Determine the matrix representation of the linear transformation $\mathscr{D} \in L(\mathcal{R}_n[\lambda] \to \mathcal{R}_{n-1}[\lambda])$,

$$\mathscr{D}(f(\lambda)) = \frac{d}{d\lambda} f(\lambda)$$

with respect to the bases $[1 \ \lambda \ \cdots \ \lambda^n]$ and $[1 \ \lambda \ \cdots \ \lambda^{n-1}]$.

Example 1.27 (p. 26). Define the linear transformation $\mathscr{S} \in L(\mathcal{R}_n[\lambda] \to \mathcal{R}_{n+1}[\lambda])$ as

$$\mathscr{S}(f(\lambda)) = \int_0^\lambda f(\xi)\, d\xi \qquad \forall\, f(\lambda) \in \mathcal{R}_n[\lambda].$$

Determine the matrix representation with respect to the basis $[1 \ \lambda \ \cdots \ \lambda^n]$ and $[1 \ \lambda \ \cdots \ \lambda^{n+1}]$.

Problem 4

Denote by $\mathbb{R}_{n-1}(\lambda)$ the set of all polynomials with real coefficients and degrees not exceeding $n-1$. With respect to the following addition and scalar product operations,

$$f_1(\lambda) = \sum_{i=0}^{n-1} f_i^{(1)} \lambda^i, \qquad f_2(\lambda) = \sum_{i=0}^{n-1} f_i^{(2)} \lambda^i$$

$$(f_1 + f_2)(\lambda) = \sum_{i=0}^{n-1} \big(f_i^{(1)} + f_i^{(2)}\big) \lambda^i$$

$$(k f_1)(\lambda) = k \sum_{i=0}^{n-1} f_i^{(1)} \lambda^i$$

(1) Let a linear transformation $\mathfrak{L}_\alpha : \mathbb{R}_{n-1}(\lambda) \to \mathbb{R}_{n-1}(\lambda)$ be defined as

$$\mathfrak{L}_\alpha[f(\lambda + \alpha)] = f(\lambda), \qquad f(\lambda + \alpha),\ f(\lambda) \in \mathbb{R}_{n-1}(\lambda),$$

where α is real constant number, show that $\mathfrak{L}_\alpha$ is a linear transformation;

(2) Find out the matrix representation L for the linear transformation $\mathfrak{L}_\alpha$ with respect to the basis $1, \lambda + \alpha, (\lambda + \alpha)^2, \cdots, (\lambda + \alpha)^{n-1}$, i.e., the matrix L satisfying

$$\mathfrak{L}_\alpha\big([1, \lambda + \alpha, (\lambda + \alpha)^2, \cdots, (\lambda + \alpha)^{n-1}]\big) \triangleq [1, \lambda + \alpha, (\lambda + \alpha)^2, \cdots, (\lambda + \alpha)^{n-1}]L.$$